Ekansh Bajpai

Robotics researcher: event-based vision, SLAM and robot learning

Paderborn, Germany · ekansh.bajpai712@gmail.com · ekansh.is-a.dev · ekansh-bajpai · ekanshbajpai

Research Interests

Event-based and neuromorphic vision for visual odometry, SLAM and scene understanding; perception for robot manipulation in dynamic scenes; learning-based robot control, from imitation learning with vision-language-action models to whole-body control.

Education

M.Sc. Electrical Systems Engineering — Paderborn University, Germany

*Thesis: Exploring Neuromorphic Front-Ends for Event-Based SLAM (**grade 1.0**)*

- Focus: signal and information processing, SLAM, neural networks (CNNs, SNNs), event-based vision

B.Tech. Electronics and Communication Engineering — 2015 – 2019

Madan Mohan Malaviya University of Technology, India

- Thesis: self-balancing object-tracking robot (PID-controlled inverted pendulum, IMU fusion, OpenCV tracking)

IEEE CIS & COLOURS Summer School: *Foundational Models Meet Robotics: From Language and Vision to Action* — Jul 2026

Paderborn University, Germany

Research Experience

Master's Thesis Research — 2026

Automatic Control Group (RAT), Paderborn University

Supervisor: Dr. Alper Yegenoglu · Examiners: Prof. Dr. Erdal Kayacan, Prof. Dr. Sina Saravi

- Replaced the CNN patch selector of the learned event-based VO pipeline DEVO with spiking neural networks (surrogate gradients); compared voxel-grid and time-surface encodings against CNN baselines in 150 runs
- MVSEC: ∼**2**× **lower absolute trajectory error** and **10**× **smaller spread** than CNN baselines; held-out TartanEvents: both viable spiking variants beat both CNNs
- Showed with a same-data CNN control that the gap on fast drone-racing motion (UZH-FPV) comes from training data, not spiking computation
- Generated **TartanEvents**, a 1.3 TB simulated event-camera dataset (open release after paper submission; access on request)
- Built SuperEventSLAM, a graph-based SLAM back-end for DEVO (keyframes, SuperEvent features, three-stage loop verification, g2o); runs end to end at 36.7 FPS, loop verification still being tuned

Master's Project *Rescue Robots*: Autonomous Mapping and SLAM Optimization

GET Lab (Prof. Dr.-Ing. Bärbel Mertsching), Paderborn University

- LOAM-based lidar SLAM with IMU pose estimation; templated C++ architecture to benchmark pose-graph back-ends (g2o, SE-Sync) in ROS

Work Experience

Student Assistant (SHK), Humanoid Robotics — Sep 2026 – present

Stratosfare – Fraunhofer IEM, Paderborn

- Implemented NVIDIA's GR00T end-to-end workflow on a real **Unitree G1** humanoid: teleoperated demonstrations, LeRobot dataset export, GR00T N1.7 VLA fine-tuning, LEAPP export and deployment on NVIDIA Jetson AGX Thor
- Set up and integrated NVIDIA GR00T WholeBodyControl (GEAR-SONIC): robot interfaces, software and runtime environment; work directly on real humanoid hardware for teleoperation and whole-body motion control
- Built the data-collection pipeline for learned whole-body control; collect whole-body VLA training data with PICO VR teleoperation
- Demonstrated the trained G1 picking an object and placing it in a container at a 3-day workshop; guided attendees through teleoperation as used for episode collection

Student Assistant, AI/ML and Software, Fraunhofer IEM: LLM-based knowledge and compliance platforms (RAG, FastAPI) — 2024 – present

Robotics Engineering Intern, Cellgo GmbH: embedded I2C/SPI for a carrier robot — Aug 2022

Teaching

Student Assistant, Nonlinear Control of Autonomous and Robotic Systems — Apr – Sep 2026

Automatic Control Group (RAT), Paderborn University

- Built the ROS 2/Gazebo quadrotor framework and tutorials (PID, feedback linearization, sliding mode, minimum-snap); supported Lyapunov stability analysis; graded term projects

Publications, Datasets and Software

- E. Bajpai. *Exploring Neuromorphic Front-Ends for Event-Based SLAM*. Master's thesis, Paderborn University, 2026. Conference paper with my supervisor: in preparation.
- **TartanEvents** (dataset, 1.3 TB): huggingface.co/datasets/upb-rat-ekansh/TartanEventsLeft_wo_upsampling
- **devo-vo**: portable subset of DEVO for event-based VO (github.com/ekansh-bajpai/devo-vo); **pangolin-python-viz**: Python visualisation library on Pangolin for SLAM (github.com/ekansh-bajpai/pangolin-python-viz); snn-devo, supereventslam: private, access on request

Skills

Perception	event cameras, spiking neural networks (snnTorch), visual odometry, SLAM, pose-graph optimization, bundle adjustment, OpenCV
Robot learning	imitation learning, VLA fine-tuning (GR00T, LeRobot), teleoperation, whole-body control, Unitree G1
Control	nonlinear control, sliding mode, feedback linearization, Lyapunov stability, PID, trajectory generation
Programming	Python, C++, PyTorch, ROS 2, Eigen, g2o, Gazebo, Git, Linux

References: Prof. Dr. Erdal Kayacan (Chair, RAT) · Dr. Alper Yegenoglu (thesis supervisor, RAT), Paderborn University